

Supervised Neural Network Project: GNN-Based BERT for Understanding Context from Music

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Abstract—This work presents an experimental study of GNN-based audio representations and BERT-based text representations for music-context understanding. Audio clips are modeled as temporal graphs using segment-level acoustic features, while captions are encoded using DistilBERT. The study evaluates text classification, graph-based audio modeling, GNN-BERT fusion, and contrastive graph-text retrieval. Results show that text-based modeling outperforms the current graph representation, and frozen multimodal fusion does not improve Macro-F1 over the BERT-only baseline. Contrastive training improves exact-pair retrieval, although a pretrained zero-shot CLAP model performs relatively better. These findings show the limitations of frozen encoders and coarse graph representations and motivate future work on fine-tuning audio modeling.

Index Terms—Music Understanding, GNN, BERT, CNN, RNN, Multimodal Learning, Audio-Text Retrieval, Graph-Text Fusion, Contrastive Learning.

I. INTRODUCTION

Music is a complex form of information that contains multiple layers of context, including melody, harmony, rhythm, lyrics, genre, mood, and other characteristics. Therefore, understanding music requires models that can capture both the acoustic properties of a track and the relationships between its different components. The project specification describes music context as a combination of audio structure, textual information, and semantic labels such as genre and emotion. Traditional approaches to music understanding commonly represent audio as sequential features such as mel-spectrograms or chroma representations and process them using convolutional or recurrent neural networks. These approaches fail to learn relational structures that are not naturally represented as simple sequences.

The proposed project therefore considers a music track as a combination of audio features, textual information, and a structure graph, where graph vertices represent musical segments, chords and edges represent transitions or similarities.

Transformer-based language models such as BERT can learn contextual representations from textual inputs, making them useful for understanding lyrics, music tags, and natural-language descriptions. In the project, BERT is used to trans-

form textual music information into contextual embeddings that can subsequently be used for music-context prediction.

Graph Neural Networks (GNNs) are designed to learn from graph-structured data. GNN can incorporate information about both individual musical elements and their relationships. The project specifically proposes Graph Attention Network (GAT) models for processing music segments or chord graphs and generating graph-level representations for classification.

BERT focuses on the semantic information contained in text, while a GNN focuses on the structural relationships represented by the music graph. This provides the basis for a hybrid GNN-BERT architecture capable of combining semantic and structural information.

II. PROBLEM STATEMENT

Understanding musical context from a single modality is challenging because different aspects of music are represented in different forms. Audio contains information about acoustic and structural characteristics, while lyrics, tags, and captions provide high-level semantic information. A model that processes only audio may fail to exploit important textual context, while a text-only model cannot directly represent relationships among musical segments or chords.

The project formulates a music track as

$$T = (X_{\text{audio}}, X_{\text{text}}, G, y)$$

where X_{audio} represents audio features, X_{text} represents lyrics, tags, or captions, $G=(V,E)$ represents the music structure graph, and (y) represents context labels such as genre, mood, or emotion. The central problem addressed in this project is: **How can textual semantic information and graph-based musical structural information be combined to improve music-context understanding and prediction?** The proposed system addresses this problem by using BERT to encode textual information and a GNN to encode the structure of the music graph. Their representations are subsequently fused and used for tasks such as multi-label music tagging and, where applicable, emotion prediction.

III. RESEARCH OBJECTIVES

The main objectives of this project are:

- 1) To develop a BERT-based model for understanding textual music context and predicting music tags.
- 2) To construct graph representations of musical segments or chord relationships from audio features.
- 3) To develop a GNN-based model for learning structural representations of music.
- 4) To combine GNN and BERT representations into a unified model for multi-context music understanding.
- 5) To compare the proposed model with CNN, BERT-only, and GNN-only baselines.

IV. DATASET AND PREPROCESSING

The proposed GNN-BERT framework requires both acoustic and textual information in order to learn complementary representations of musical content. Therefore, the project uses at least one audio-based music dataset together with a corresponding source of textual or tag-based information. The project specification recommends several datasets for this purpose, including the Free Music Archive (FMA), MagnaTagATune, DEAM, MusicCaps, Million Song Dataset with tag information, and Lakh MIDI. These datasets provide different types of information, including genre labels, music tags, emotional annotations, and natural-language descriptions.

A. Dataset and Label Construction

The experiments were conducted using metadata and captions from the MusicCaps dataset. The complete metadata contains 5,521 music-caption pairs. For the multi-label prediction task, the target labels were obtained from the independently provided AudioSet positive labels rather than being generated from caption keywords. The 30 most frequent non-constant labels were selected using training-set label frequencies, subject to a minimum of 20 positive training examples. All samples were retained, including samples with no positive label among the selected classes.

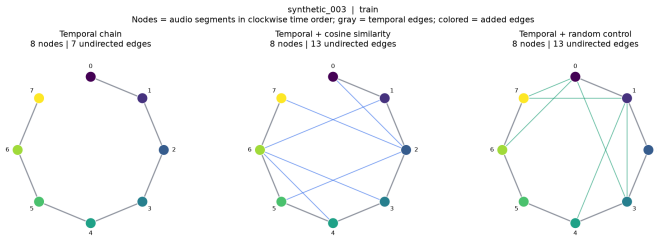


Fig. 1. Overview of the proposed framework.

For Task 1, the complete metadata was divided into training, validation, and test sets using a fixed video-ID-based split with a 70/15/15 ratio and random seed 42, resulting in 3,864 training, 828 validation, and 829 test samples. Tasks 2–4 used the subset of samples for which the corresponding audio files were successfully available and passed the audio validation checks. This resulted in 3,964 usable clips, divided into

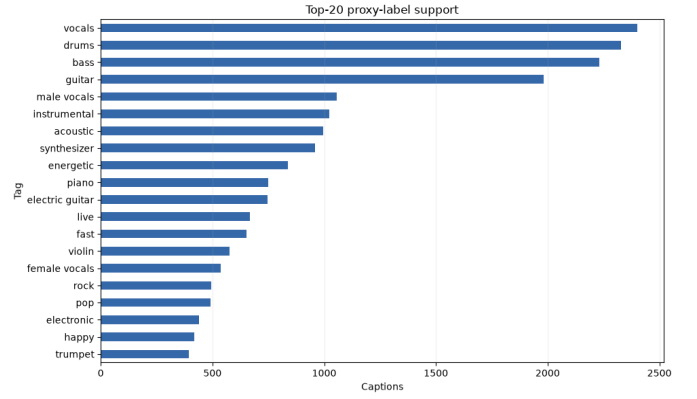


Fig. 2. Proxy-Label

2,775 training, 583 validation, and 606 test samples using the inherited split.

B. Audio Preprocessing and Feature Extraction

For the graph-based experiments, each audio clip was cropped to its annotated time interval, converted to mono, resampled to 22.05 kHz, and peak-normalized. Each clip was then divided into ten non-overlapping one-second segments. Every segment was represented by a 311-dimensional feature vector consisting of:

- 128 log-mel spectrogram mean features,
- 128 log-mel spectrogram standard-deviation features,
- 12 chroma mean features,
- 20 MFCC mean features,
- 20 MFCC-delta mean features,
- Mean RMS energy,
- Spectral centroid, and
- Zero-crossing rate.

Feature normalization was fitted using training-set nodes only and then applied to the validation and test nodes to avoid information leakage.

C. Graph Construction

Each audio clip was represented as a graph with ten nodes, where each node corresponds to one one-second audio segment. The primary temporal graph connected adjacent segments, producing a chain of nine undirected edges. To evaluate the effect of alternative connectivity patterns, two additional graph variants were constructed.

The similarity graph included the temporal edges together with symmetrized top-2 cosine-similarity connections between non-adjacent segments. The random-control graph retained the temporal chain and added the same number of randomly selected non-adjacent edges. These graph variants allowed the experiments to compare temporal structure, feature-based similarity, and random connectivity under approximately matched edge counts.

V. PROPOSED METHOD

The proposed framework investigates how textual and audio-structural information can be combined for music understanding. The complete methodology consists of four related stages: text-based multi-label classification, graph-based audio modeling, multimodal GNN-BERT fusion, and contrastive graph-text retrieval.

A. Text-Based Modeling

The textual baseline uses `distilbert-base-uncased` to encode MusicCaps captions. The model uses the contextual representation of the `[CLS]` token, followed by dropout and a linear classification layer with 30 sigmoid outputs. Binary cross-entropy with logits is used for multi-label prediction. This model provides the text-only baseline and its fine-tuned encoder is later reused as a frozen text representation in the multimodal experiments.

B. Graph-Based Audio Modeling

Each audio clip is represented as a graph containing ten segment nodes. A two-layer GraphSAGE encoder with mean aggregation is applied to the node features. Each layer uses a hidden width of 128 with LayerNorm, ReLU activation, and dropout. The final graph representation is obtained by concatenating global mean and maximum pooling over the node embeddings, producing a clip-level audio representation.

Three graph connectivity settings are evaluated: temporal adjacency, temporal adjacency with feature-similarity edges, and temporal adjacency with randomly selected control edges. A pooled-feature MLP is also used to determine whether any performance improvement is caused by graph message passing or simply by using the same acoustic features.

C. Multimodal GNN-BERT Fusion

The multimodal model combines the frozen text representation from the fine-tuned DistilBERT encoder with the frozen temporal-graph GraphSAGE representation. Both representations are projected into a common lower-dimensional space before classification. Three fusion mechanisms are investigated:

- **Concatenation:** concatenates the projected text and graph representations.
- **Gated fusion:** learns a sigmoid-based gate to control the contribution of each modality.
- **Cross-attention:** uses the graph representation as a query over the contextual text token representations.

Each fusion head is followed by dropout and a 30-label linear prediction layer. BERT-only and GNN-only models are included as unimodal controls.

D. Contrastive Graph-Text Retrieval

To evaluate cross-modal alignment beyond classification, a dual-encoder retrieval model is trained using the frozen text and graph representations. Separate linear projection layers map the 768-dimensional text representation and

256-dimensional graph representation into normalized 64-dimensional embeddings. A symmetric InfoNCE contrastive objective is used, where the matching audio-caption pair is treated as the positive example and other samples in the batch serve as negatives.

Retrieval performance is evaluated in both directions: text-to-graph and graph-to-text. Recall@1, Recall@5, Recall@10, median rank, and mean reciprocal rank are reported on the held-out test pairs.

VI. EXPERIMENTAL SETUP

All experiments were implemented using PyTorch and conducted on a CPU-based environment using six computational threads. The same training, validation, and test partitions were maintained across the experiments to ensure consistent comparisons. The graph-based and multimodal experiments used the 3,964 clips with successfully validated audio, consisting of 2,775 training, 583 validation, and 606 test samples.

TABLE I
TRAINING CONFIGURATIONS FOR THE IMPLEMENTED MODELS.

Model	Batch	Epochs	Learning Rate	Weight Decay
DistilBERT classifier	16	2	$2 \times 10^{-5} / 10^{-3}$	10^{-5}
GraphSAGE models	16	30	10^{-3}	10^{-5}
Fusion heads	16	20	10^{-3}	10^{-2}
Contrastive projections	32	30	10^{-3}	10^{-2}

A. Training Configuration

The text classifier was fine-tuned for two epochs using AdamW, with a learning rate of 2×10^{-5} for the DistilBERT encoder and 10^{-3} for the classification head. The maximum token length was 128 and the batch size was 16. A binary cross-entropy loss with logits was used for multi-label classification.

The graph models were trained for up to 30 epochs with a batch size of 16, learning rate of 10^{-3} , weight decay of 10^{-5} , and dropout of 0.2. Early stopping with a patience of five epochs was applied, and the best checkpoint was selected using validation Macro-F1. The multimodal fusion heads were trained for up to 20 epochs using AdamW with a learning rate of 10^{-3} , weight decay of 10^{-2} , batch size 16, and dropout of 0.1. The text and graph encoders were frozen during fusion training.

For contrastive retrieval, the projection heads were trained using a symmetric InfoNCE loss with temperature 0.07, batch size 32, learning rate of 10^{-3} , and weight decay of 10^{-2} . The text and graph encoders were also frozen in this experiment.

B. Evaluation Protocol

Multi-label classification performance was evaluated using Macro-F1, Micro-F1, and mean Average Precision (mAP). Classification thresholds were selected using the validation set over a grid from 0.05 to 0.95, with no test-set-based threshold selection.

The graph and multimodal experiments were repeated using random seeds 42, 43, and 44, and results are reported as mean $\pm$ sample standard deviation. The retrieval experiment used the same three seeds and evaluated exact audio–caption pair matching in both text-to-graph and graph-to-text directions using Recall@1, Recall@5, Recall@10, median rank, and mean reciprocal rank.

VII. RESULTS

A. Overall Experimental Results

The experiments evaluated text-only, graph-based, multimodal, and contrastive graph–text approaches. The classification results show that the text representation was generally stronger than the audio graph representation for the selected AudioSet labels. The main results are summarized in Table II.

TABLE II
SUMMARY OF THE MAIN EXPERIMENTAL RESULTS.

Model	Macro-F1	Micro-F1	mAP
DistilBERT (Task 1)	0.3840	0.5902	0.3686
Pooled-feature MLP	0.2610	0.5595	0.2747
Temporal GraphSAGE	0.2648	0.5729	0.2762
BERT-only fusion control	0.3850	0.6564	0.3892
Concatenation fusion	0.3839	0.6560	0.3994
Gated fusion	0.3705	0.6466	0.3995
Cross-attention fusion	0.3665	0.6453	0.3861

B. Task 1: Text-Based Classification

The fine-tuned DistilBERT classifier achieved a test Macro-F1 of 0.3840, Micro-F1 of 0.5902, and mAP of 0.3686. It improved Macro-F1 and mAP over the label-prior and ontology-keyword baselines. However, the TF-IDF logistic regression baseline achieved a higher Micro-F1 of 0.6530, indicating that simple lexical features were effective for frequent labels.

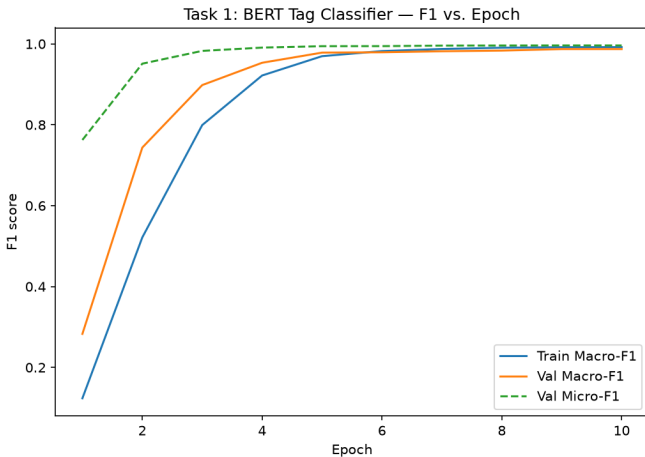


Fig. 3. Macro-F1 and Micro-F1 VS EPOCH.

The difference between Macro-F1 and Micro-F1 suggests that performance was uneven across labels. Frequent labels, particularly the “Music” category, contributed strongly

to aggregate micro-level performance. Validation-based per-label threshold calibration improved macro-level balance but reduced Micro-F1, showing that threshold selection has a considerable effect under label imbalance.

C. Task 2: Graph-Based Audio Modeling

The temporal GraphSAGE model achieved a Macro-F1 of 0.2648 ± 0.0114 , Micro-F1 of 0.5729 ± 0.0207 , and mAP of 0.2762 ± 0.0006 . It slightly outperformed the pooled-feature MLP, whose Macro-F1 was 0.2610 ± 0.0097 . This indicates that temporal message passing provided a small benefit beyond global feature pooling.

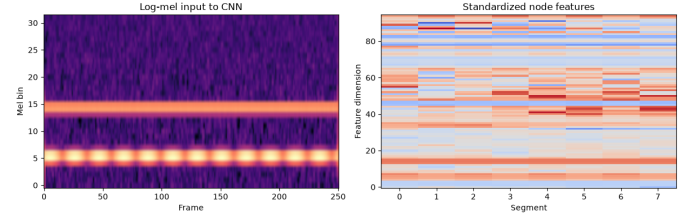


Fig. 4. Log-mel input to CNN.

However, the improvement was limited. The audio CNN performed worse, with a Macro-F1 of 0.2215 ± 0.0104 . Adding feature-similarity edges also did not improve performance: the temporal-plus-similarity GraphSAGE model achieved a Macro-F1 of 0.2540 ± 0.0145 . The random-edge control performed even lower, with a Macro-F1 of 0.2475 ± 0.0140 . These results suggest that the simple temporal chain was more useful than either similarity-based or randomly added connectivity for this feature representation.

D. Task 3: GNN–BERT Fusion

Task 3 evaluated whether audio graph representations could improve the text-based classifier. The experiments used frozen encoders: the fine-tuned DistilBERT representation from Task 1 and the temporal GraphSAGE representation from Task 2. Five model configurations were compared over three random seeds. The detailed results are shown in Table III.

TABLE III
FROZEN GNN–BERT FUSION RESULTS OVER THREE RANDOM SEEDS.
VALUES ARE MEAN $\pm$ SAMPLE STANDARD DEVIATION.

Model	Val Macro-F1	Test Macro-F1	Test Micro-F1
BERT-only	0.4621 ± 0.0040	0.3850 ± 0.0058	0.6564 ± 0.0105
GNN-only	0.2967 ± 0.0050	0.2539 ± 0.0119	0.5632 ± 0.0352
Concatenation	0.4228 ± 0.0070	0.3839 ± 0.0124	0.6560 ± 0.0034
Gated fusion	0.4353 ± 0.0048	0.3705 ± 0.0013	0.6466 ± 0.0149
Cross-attention	0.4232 ± 0.0058	0.3665 ± 0.0102	0.6453 ± 0.0061

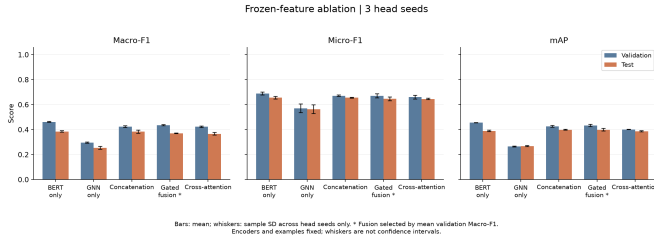


Fig. 5. Frozen-feature Ablation.

The BERT-only model achieved the best validation Macro-F1 of 0.4621 ± 0.0040 and the best test Macro-F1 of 0.3850 ± 0.0058 . None of the fusion mechanisms improved test Macro-F1 over the text-only control. Concatenation was the closest, obtaining a test Macro-F1 of 0.3839, while gated fusion and cross-attention achieved 0.3705 and 0.3665, respectively.

Although gated fusion improved substantially over the GNN-only model by approximately 0.1166 Macro-F1 points, it still remained below the BERT-only model by approximately 0.0145 points. This indicates that the audio graph contained useful information, but the frozen audio representation was not sufficiently complementary to the text representation to produce an overall classification improvement.

An interesting difference appears in mAP. Gated fusion achieved an mAP of 0.3995 and concatenation achieved 0.3994, both slightly higher than the BERT-only mAP of 0.3892. Therefore, fusion improved ranking quality for some labels even though its thresholded F1 performance was lower. This suggests that the multimodal representations may contain useful information that was not fully converted into binary classification decisions by the selected thresholds.

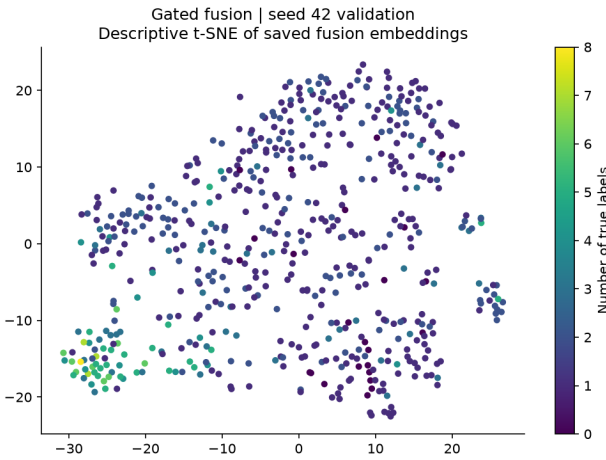


Fig. 6. Descriptive t-SNE of saved fusion Embeddings.png t-SNE.

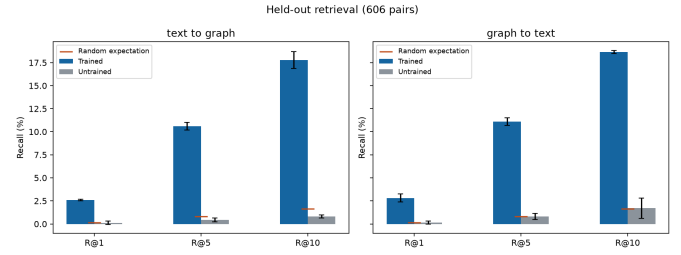


Fig. 7. Held-out Comparison.

TABLE IV
DETAILED TEST RESULTS FOR THE FROZEN FUSION MODELS.

Model	Test mAP	Trainable Parameters
BERT-only	0.3892	51,166
GNN-only	0.2680	18,398
Concatenation	0.3994	69,534
Gated fusion	0.3995	75,870
Cross-attention	0.3861	84,254

The per-example analysis further showed mixed behavior. Compared with BERT-only, the selected gated fusion model improved predictions for 438 seed/example pairs, worsened them for 452 pairs, and produced ties for 928 pairs. Thus, fusion changed individual predictions but did not provide a consistent improvement across the test set.

The qualitative sensitivity analysis also showed that graph nodes could affect predictions for different musical examples. For example, removing a one-second node from a guitar tutorial changed the predicted score for the focal guitar-related label by 0.1112. Similar changes were observed for choir and ambient synthesizer examples. These observations indicate that temporal segments influence the fusion model, although they should be interpreted as sensitivity measurements rather than causal evidence of feature importance.

E. Task 4: Contrastive Graph-Text Retrieval

The contrastive retrieval experiment evaluated whether the learned graph and text representations could be aligned in a shared embedding space. The experiment used the same 3,964-clip available-audio cohort and evaluated 606 held-out audio-caption pairs. The text and graph encoders were frozen, and only two linear projection heads were trained using a symmetric InfoNCE objective. Therefore, the reported result represents projection-only contrastive training over previously supervised representations rather than joint fine-tuning of the encoders.

As shown in Table V, contrastive training substantially improved retrieval over untrained projection heads in both directions. Text-to-graph Recall@10 increased from 0.83% to 17.77%, while graph-to-text Recall@10 increased from 1.71% to 18.65%. Nevertheless, most exact paired items were not retrieved within the top ten candidates, indicating that the alignment quality remained limited.

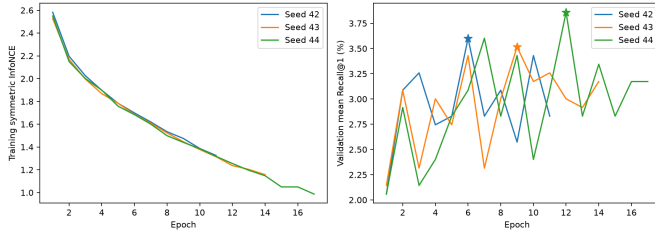


Fig. 8. Training Validation.

TABLE V

FROZEN GRAPH-TEXT RETRIEVAL RESULTS ON 606 HELD-OUT PAIRS. VALUES ARE MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER THREE SEEDS.

Direction	Model	R@1	R@5	R@10	Med. Rank	MRR
Text $\rightarrow$ Graph	Trained	2.59 $\pm$ 0.10	10.62 $\pm$ 0.42	17.77 $\pm$ 0.91	52.17	0.0800
Text $\rightarrow$ Graph	Untrained	0.11 $\pm$ 0.19	0.44 $\pm$ 0.19	0.83 $\pm$ 0.17	314.83	0.0091
Graph $\rightarrow$ Text	Trained	2.81 $\pm$ 0.44	11.11 $\pm$ 0.42	18.65 $\pm$ 0.17	50.67	0.0829
Graph $\rightarrow$ Text	Untrained	0.17 $\pm$ 0.17	0.83 $\pm$ 0.33	1.71 $\pm$ 1.10	321.83	0.0112

1) *Comparison with Pretrained CLAP*: A fixed pretrained CLAP checkpoint, laion/clap-htsat-unfused, was evaluated as an external zero-shot audio-text comparator. The model was not trained or tuned on the project data, and the same 606 test IDs and captions were used for comparison. Unlike the proposed graph system, CLAP directly produces audio-text embeddings and therefore does not isolate the contribution of graph structure.

TABLE VI

COMPARISON BETWEEN THE PROPOSED FROZEN GRAPH RETRIEVAL SYSTEM AND ZERO-SHOT CLAP RETRIEVAL.

Direction	System	R@1	R@5	R@10	Med. Rank	MRR
Caption $\rightarrow$ Audio	Graph projections	2.59	10.62	17.77	52.17	0.0800
Caption $\rightarrow$ Audio	CLAP zero-shot	13.70	32.67	45.38	13	0.2380
Audio $\rightarrow$ Caption	Graph projections	2.81	11.11	18.65	50.67	0.0829
Audio $\rightarrow$ Caption	CLAP zero-shot	11.39	31.52	40.92	16	0.2178

CLAP substantially outperformed the current graph-based representation, achieving Recall@10 values of 45.38% for caption-to-audio retrieval and 40.92% for audio-to-caption retrieval. This result indicates that the current graph representation is considerably weaker than a large pretrained audio-text model for exact-pair retrieval. However, this comparison changes both the audio-text backbone and the source of pretraining, so it cannot be interpreted as a controlled ablation of graph structure. Possible overlap between CLAP’s external pretraining data and the evaluated clips was not ruled out.

2) *Zero-Shot Caption Tag Prediction*: A separate zero-shot caption-based baseline was evaluated using CLAP text embeddings and fixed positive/negative prompts for each of the 30 AudioSet labels. The score was computed from the cosine-similarity margin between the positive and negative prompts, without local training, prompt search, or validation calibration.

TABLE VII
ZERO-SHOT CAPTION TAG PREDICTION COMPARED WITH SUPERVISED GATED FUSION.

Method	Macro-F1	Micro-F1	mAP
Zero-shot caption prompt similarity	0.1090	0.1530	0.1025
Supervised gated fusion	0.3705	0.6466	0.3995

The zero-shot caption prompt baseline performed poorly compared with the supervised gated fusion model. However, this result should be interpreted narrowly: the method is a prompt-similarity baseline rather than a supervised natural-language-inference classifier or an audio-based tag classifier. Its low performance may reflect the difficulty of distinguishing positive and negated textual prompts using the fixed CLAP text representation.

3) *Human Evaluation Status and Limitations*: A listening-study package was prepared using ten fixed caption queries, with the top-three retrieved clips from the graph system and CLAP. The planned evaluation assigns five independent listeners to rate caption-clip relevance on a 1–5 scale. However, no human ratings have yet been collected. Therefore, the current study reports only automatic exact-ID retrieval metrics and does not make claims about subjective semantic relevance or perceived audio-caption matching quality.

The main limitation of Task 4 is that the current implementation trains only projection heads while keeping the encoders frozen. This differs from a strict joint-fine-tuning interpretation of the original contrastive-learning objective. In addition, the evaluation is based on exact paired IDs, meaning that semantically relevant but non-identical clips are counted as incorrect. Other limitations include the available-audio selection bias, coarse one-second graph features, classification-trained representations, lack of artist-disjoint evaluation, and the relatively small retrieval gallery.

F. Discussion

Overall, the results demonstrate that textual information was the strongest source for the selected labels, while the graph-based audio representation provided complementary but limited information. Temporal GraphSAGE slightly outperformed feature pooling, but more complex graph connectivity did not yield consistent gains. More importantly, the frozen GNN-BERT fusion models did not outperform the BERT-only baseline in Macro-F1, suggesting that stronger audio representations, finer temporal modeling, or joint encoder fine-tuning may be required for effective multimodal improvement.

The higher mAP of the concatenation and gated fusion models suggests that multimodal information may still improve label ranking even when thresholded classification performance decreases. Therefore, the findings do not indicate that audio information is useless; rather, they show that the current frozen encoders, coarse one-second features, global graph pooling, and imbalanced labels limit the effectiveness of the proposed fusion strategies.

VIII. CONCLUSION

This work presented an experimental study of GNN-based and BERT-based approaches for understanding musical context from audio and text. The results showed that the fine-tuned DistilBERT model provided the strongest classification performance among the individual modalities, while the temporal GraphSAGE model achieved only a modest improvement over pooled acoustic features. Alternative similarity-based and random graph connections did not improve over the simple temporal graph.

The frozen GNN-BERT fusion experiments showed that multimodal integration did not improve Macro-F1 over the BERT-only baseline. Concatenation performed closest to the text-only model, while gated fusion and cross-attention produced lower F1 scores. Nevertheless, the improved mAP of the fusion models suggests that audio information may still contribute to label ranking, even when it does not improve threshold-based classification.

The contrastive retrieval experiment improved exact graph to text matching over untrained projection heads, reaching Recall@10 values of 17.77% and 18.65% in the two retrieval directions. However, a pretrained zero-shot CLAP comparator performed substantially better, while the planned human listening evaluation remains pending. These results suggest that the current frozen graph representation is not yet competitive with large-scale pretrained audio-text models for cross-modal retrieval.

The current study is limited by the available-audio subset, frozen encoders in the multimodal experiments, coarse one-second acoustic features, and evaluation based on exact metadata pair matching. Future work should investigate end-to-end multimodal training, richer audio representations, artist-disjoint evaluation, larger datasets, and human-centered assessment of semantic music understanding.